

Krishna Auto Care Precision Workshop

Full System Documentation & Detailed Role Functionality Guide

System Architecture Overview: Krishna Auto Care is a specialized web management platform designed for automotive workshops. It connects garage owners, service advisors, and mechanics in a unified workflow. Key capabilities include job card lifecycle management, automated dynamic WhatsApp customer notifications, invoice generation with vehicle pickup/drop charges, cloud photo storage (Cloudinary), AMC contracts, spare parts inventory tracking, and an admin-only Incentive & Net Profit Calculator.

1. User Roles & Permission Matrix Overview

System Role	Primary Responsibility	Access Scope	Key Modules
Garage Owner / Admin (owner)	Overall Garage Operations, Financial Governance, Staff Management	Full Read & Write across all system modules, financial reports, and settings.	All Modules + Owner Dashboard, Net Profit Calculator, Expense Logger, Staff Manager
Service Advisor (advisor)	Customer Service, Vehicle Intake, Job Cards, Billing & Invoicing	Operational Read & Write for front-desk activities. Cannot view profit/incentive pool.	Advisor Dashboard, Vehicles, Job Cards, Invoices, WhatsApp Updates, AMC Contracts
Mechanic / Technician (mechanic)	Vehicle Repair Execution, Status Logging, Parts Usage Tracking	Restricted to assigned jobs only. Cannot access financial or customer billing data.	Mechanic Dashboard, My Job Cards, Status Logger, Labour & Parts Log
Custom Staff Roles	Specialized Job Titles (e.g. Senior Electrician, Quality Inspector)	Custom designation labels configured under Staff Management by Admin.	Configured per staff profile

2. Detailed Functionalities by Role

■ Garage Owner / Admin (Role: owner)

- **Executive Owner Dashboard:** Live KPI view displaying active jobs, completed jobs, monthly revenue, pending customer invoices, total customers, vehicles, and low-stock spare part alerts.
- **Staff Account Management & Custom Roles:** Create new staff login credentials, update staff roles, and assign custom designations (e.g., *Senior Technician*, *Diagnostic Specialist*, *Parts Officer*).
- **Incentive & Net Profit Calculator (Exclusive Access):** Calculate actual garage net profit using $\text{Net Profit} = \text{Paid Invoices} - \text{Total Expenses}$. Dynamically adjust incentive percentages (5%, 10%, 15%, 20%) to view live pool distributions.
- **Per-Mechanic Performance Breakdown:** Track completed jobs, billed labor, parts cost, net profit contribution, and calculated incentive payouts per mechanic.
- **Workshop Expense Logger:** Record overhead operating expenses (Rent ■, Utilities ■, Tools ■■, Staff Advances ■, Parts Purchase ■■) with complete audit and deletion options.
- **Financial Reports & Analytics:** View date-range revenue summaries, expense breakdowns, and labor vs. parts profitability reports.
- **Inventory & Supplier Management:** Full authority to manage spare parts inventory, safety stock thresholds, unit costs, pricing, and supplier records.
- **Customer & Vehicle Master Directory:** Access, edit, or remove customer records, vehicle profiles, and service histories.

■ Service Advisor (Role: advisor)

- **Service Advisor Dashboard:** Real-time view of daily vehicle intake, active job cards, pending customer bills, and search tools.
- **Customer & Vehicle Registration:** Register new customers and vehicles (license plate, VIN, make, model, year, color, mileage). Upload vehicle photos directly to Cloudinary CDN.
- **Job Card Creation & Lifecycle Management:** Open new job cards, record customer problem descriptions, document initial vehicle condition, and assign primary mechanics.
- **Dynamic WhatsApp Customer Updates:** Send status-specific WhatsApp notifications with 1 click directly from the job card (e.g., *WAITING FOR PARTS* ■■, *IN PROGRESS* ■, *COMPLETED* ■, *DELIVERED* ■).
- **Invoice Generation & Billing:** Convert completed job cards into customer invoices. Automatically sums labor charges, parts used, and GST taxes.
- **Vehicle Pickup & Drop Charge Option:** Dynamic toggle to add custom vehicle pickup and drop-off charges directly to invoices.
- **Payment Collection Tracking:** Record invoice payment statuses (Unpaid, Partially Paid, Paid) and generate printable PDF invoices for customers.
- **Annual Maintenance Contracts (AMC):** Enroll customer vehicles in AMC plans (Silver, Gold, Platinum), track validity dates, and record free service benefits.

■ Mechanic / Technician (Role: mechanic)

- **Personalized Mechanic Dashboard:** Clean workspace displaying exclusively job cards assigned to that specific mechanic.
- **Job Execution & Status Logging:** Update real-time job status (*In Progress*, *Waiting for Parts*, *Completed*) so advisors and owners can track progress.
- **Labour & Time Logger:** Log labour hours worked and specific repair actions performed on vehicles.
- **Spare Parts Usage Logger:** Record spare parts and consumables used from inventory during vehicle repair.
- **Performance Contribution:** Completed jobs automatically feed into the admin incentive calculator to generate monthly performance rewards.

■■ Custom Staff Roles & Specializations

- **Flexible Workshop Designations:** Admins can assign custom titles (e.g. *Master Electrician*, *Denter & Painter*, *AC Service Specialist*, *Quality Inspector*) without breaking system permissions.
- **Role Display Alignment:** Custom titles appear cleanly across Staff Directory listings, Job Cards, and Customer Invoices.

3. Technical Architecture & Environment

Component	Technology / Library	Implementation Details
Backend Core	Django 5.x / Python 3.14	MVC architecture, ORM models, custom authentication backend, and RBAC decorators.
Database	PostgreSQL / SQLite	Relational data schema for Users, Profiles, Customers, Vehicles, JobCards, Invoices, Expenses, and AMCs.
Cloud Storage	Cloudinary CDN	Persistent image storage (`django-cloudinary-storage`) solving Vercel read-only filesystem limits.
Frontend UI	Vanilla CSS, Glassmorphism, SVG Icons	Responsive layout, HSL color tokens, inline SVG icons, and mobile-friendly touch cards.
Deployment	Vercel Serverless (WSGI)	Automated CI/CD deployment pipeline synced with GitHub main branch.